

Data Analytics

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Course : Data Analytics

Subject : SQL

Batch Time : 8 to 9

1. What is the functional difference between SELECT * and specifying column names, and when is each preferred?

- SELECT *
 - When we need every column
 - Get all the Data from the Table
 - EX:- SELECT * FROM superstore;
- SELECT 'Order ID', Sales, Profit FROM superstore;
 - In reports, In applications
 - Give only Specific column's Data
 - EX:- SELECT 'Order ID', Sales FROM superstore;

2. Which keyword renames a column in the output, and does this alias change the actual table structure in the database?

- 'AS' keyword used to rename the column's name
- It's not change the actual structure of the table in Database
- EX :- SELECT Sales AS Total_Sales FROM superstore;

3. Why does wrapping a numeric value in quotes (e.g., '5000') in a WHERE clause create a data type conflict in SQL?

- Suppose Sales is a numeric column.
- WHERE Sales > 5000 // Numeric Value
- WHERE Sales > '5000' // Considered as String
- It perform Slower performance
- In this there are possibility to see conversion errors
- It considered as Data type mismatch

4. Contrast the results of ORDER BY Profit DESC versus ASC when the goal is to identify the top 10 most profitable orders

- DESC = Descending
 - Largest to smallest
 - ORDER BY Profit DESC LIMIT 10;
- ASC = Ascending
 - Smallest to largest
 - ORDER BY Profit ASC LIMIT 10;

5. What is the T-SQL equivalent of the LIMIT clause in MS SQL Server, and why does syntax vary across SQL engines?

- MySQL

```
SELECT *  
FROM superstore  
LIMIT 10;
```

- SQL Server

```
SELECT TOP 10 *  
FROM superstore;
```

6 Explain the logical execution order of a query containing SELECT, WHERE, ORDER BY, and LIMIT clauses

- When we write a SQL query, it looks like SQL executes it from top to bottom, but internally it follows a different order.

- For a query containing **SELECT, WHERE, ORDER BY, and LIMIT**, the logical execution order is:

- **FROM** – SQL first reads the data from the table.
- **WHERE** – It filters the rows based on the given condition.
- **SELECT** – It chooses the columns that need to be displayed.
- **ORDER BY** – It sorts the result in ascending or descending order.
- **LIMIT** – It returns only the specified number of rows.